

Project Proposal

TIH - IoT CHANAKYA Fellowship Program 2025

CreditMitra: India's Privacy-Preserving Agentic AI System for Automated Credit Underwriting from Heterogeneous Bank Statements

Submitted by

Dr. Sandesh Jain

Assistant Professor, ABV-IIITM Gwalior

18 December 2025



TIH Foundation for IoT & IoE
3rd Floor, IITB-Monash Research Academy Building,
IIT Bombay Campus, Mumbai 400076, Maharashtra, India
office.tih@iitb.ac.in | www.tih.iitb.ac.in
CIN: U73200MH2020NPL344467



TIH reserves all rights in this document and in the information contained therein. Reproduction, use or disclosure to third parties without express authority is strictly forbidden.

Version 1.0.0
28 November-2025
TIH

1 Team details

Sr. No.	Name	Details
1	Project Title	CreditMitra: India's Privacy-Preserving Agentic AI System for Automated Credit Underwriting from Heterogeneous Bank Statements
2	Faculty Mentor	Dr. Sandesh Jain (PI/Mentor) and Dr. Somesh Kumar (Co-PI/Co-Guide)
	Name & Dept.	Electrical and Electronics Engineering
	College Name	ABV-IIITM Gwalior
3	Student details	
a	Team Leader (1) Name	Disha Kwatra
	Degree/Dept/Year	IMG (Integrated B. Tech + MBA), Information Technology Department, 3rd year
b	Member 2 Name	Gargi Kalia
	Degree/Dept/Year	IMT (Integrated B. Tech + M. Tech), Information Technology Department, 2nd year
c	Member 3 Name	Shivansh Katiyar
	Degree/Dept/Year	IMG (Integrated B. Tech + MBA), Information Technology Department, 2nd year

Duration of the project: UG students – up to 10 months; PG students - up to 2 years

2 Executive Summary

Bank statements in India lack a standardized format, making information extraction from heterogeneous layouts difficult and forcing lenders to rely on manual reviews or expensive third-party platforms such as Perfios, which also introduce privacy risks. Existing credit-risk models are largely static and fail to capture real-time transaction behavior as customer

financial patterns evolve. Payee-level details are especially critical; for example, frequent payments to gambling platforms may indicate higher default risk compared to stable, consumption-linked spending. These challenges highlight the need for a scalable, privacy-preserving AI system that can accurately extract transactions, identify payees, and integrate these signals into dynamic credit-risk models. To enable nationwide deployment, the system should leverage a distributed LLM architecture, where lightweight models run on client devices and heavier components operate on backend nodes, reducing latency and ensuring sensitive financial text remains close to the source.

3 Statement of the problem

Indian banks do not adhere to a unified format for bank statements. They employ a variety of layouts, table structures, narration styles, and field conventions. This heterogeneity in the representation of transaction strings, combined with often noisy and unstructured transaction descriptions, makes automated credit assessment a significant challenge [1].

As a result, lenders often remain reliant on manual reviews, rule-based parsing techniques, and costly third-party orchestrators. This dependency not only delays the loan processing timeline but also hinders scalable underwriting operations. Furthermore, because banks rely on these third-party applications, there are inherent privacy risks and additional costs involved. Additionally, most credit-risk models remain static or are only periodically updated, which makes it difficult for them to adapt to the growing volume and complexity of business transactions.

Understanding the payee information within transaction strings is crucial for credit-risk modeling. Although many existing credit models utilize this information, they often remain static or only update periodically [2]. This static nature limits their ability to keep pace with the increasing volume and complexity of business transactions.

One key challenge is that non-merchant transactions typically offer little to no added value for risk assessment. Moreover, there can be multiple payees with the same name, which complicates the process of identifying merchant-related transactions.

In our simulation studies, we discovered that only about 10% of transactions are merchant transactions. This makes it essential to classify payees into merchant and non-merchant categories. To achieve this, we use a fine-tuned SLM or a trained classifier, with an emphasis on high recall to ensure that we accurately identify as many merchant payees as possible.

Our model is designed to be lightweight so that it can be easily deployed in local computing environments. Instead of relying on costly and resource-intensive Google searches to find payee information, we leverage a Large Language Model's knowledge base for known merchants. As a fallback, we use Langsearch [3], a retrieval-augmented generation (RAG) [4] system based on internet data, which is both fast and cost-effective while providing accurate information.

Additionally, most of the machine learning models have issues of data leakage on Inference, this problem also exists with credit risk models. To address this challenge, we introduce an on-device differential privacy-enhanced small language model (DP-SLM) [5], inspired by the differential privacy framework for LLMs (DPF-LLM) [6]. In addition, we propose a distributed LLM architecture in which lightweight models operate collaboratively across

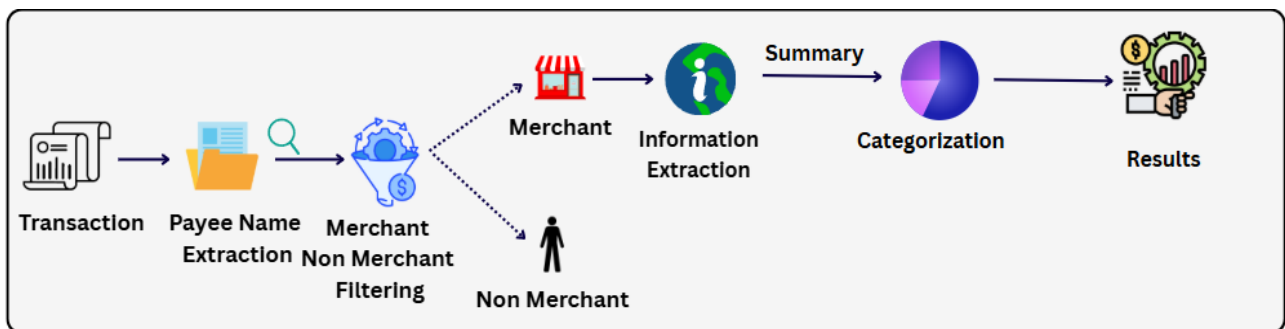


Fig. 1: Basic Pipeline of CreditMitra

clients' devices and secure backend nodes, enabling privacy-preserving and scalable inference. This design improves inference latency, reduces central computing load, and ensures that sensitive information is processed as close to the source as possible, supporting large-scale parallel processing. Sensitive financial text, such as narration details, UTRs, and salary information, is processed locally or inside confidential environments, preventing exposure of raw data. **Differentially private training (DP-SGD)** further protects

against leakage, ensuring compliance with **RBI data localization norms, GDPR, and SOC2**. Basic pipeline of CreditMitra is shown in Fig. 1.

4 Objectives / Solution

1. **Efficient extraction of Bank transaction information from heterogeneous bank statements.**
Given the heterogeneity of Indian bank statements, the system must reliably extract the maximum amount of relevant transaction information while effectively managing differences in formatting, structure, and presentation.
2. **Efficient extraction of payee name and merchant/non merchant classification.**
Payee names in transaction strings provide important insight into a borrower's financial behavior. Since banks use different transaction-string formats, accurate extraction of payee names is essential. Because most transactions are non-merchant and add limited value, the system must also classify merchant vs. non-merchant transactions with very high recall for merchant detection.
3. **Making Credit risk detection models dynamic by adding real time merchant information.**
Most credit-risk models are static or updated only periodically, making them unable to capture newly emerging merchants. Therefore, incorporating real-time merchant information is essential to keep the models dynamic and responsive.
4. **Ensure user privacy throughout the app, and credit risk models.**
Financial data is highly confidential, and credit-risk models can leak training information during inference. Therefore, strong privacy protections must be integrated throughout the system's design and development.
5. **Store resolved merchant information for faster processing in the future**
After a merchant is identified once, their details are stored in a local merchant database. This avoids repeated LLM or Langsearch calls for the same merchant, improving response time and reducing computational costs.
6. **Introduce a distributed LLM architecture for security and low latency**
To ensure scalability, a distributed computing architecture must be implemented, and a distributed LLM setup built on this paradigm should be used.

5 Product / Solution Category of the proposed project

Improvements to existing Products with significant value add:

Qualifying Features: Most existing credit risk models are largely static and, even when dynamic, are updated only periodically, causing them to miss newly emerging businesses. With the rapid growth in transaction volume and business diversity, credit risk models must become more adaptive. We propose a dynamic pipeline that incorporates real-time merchant information extracted directly from transaction data. Merchant identification is performed using a Large Language Model (LLM) knowledge base with a LangSearch fallback for real-time retrieval. Transactions are categorized into 20 merchant categories, improving merchant classification accuracy from 49% to 91%. Additionally, privacy risks are mitigated by training models using Differentially Private Stochastic Gradient Descent (DP-SGD), ensuring strong privacy guarantees during training and inference. **We have already made initial prototype of CreditMitra as shown in Fig. 2, where users upload transaction data in an unstructured format, which is then automatically analyzed and categorized.**

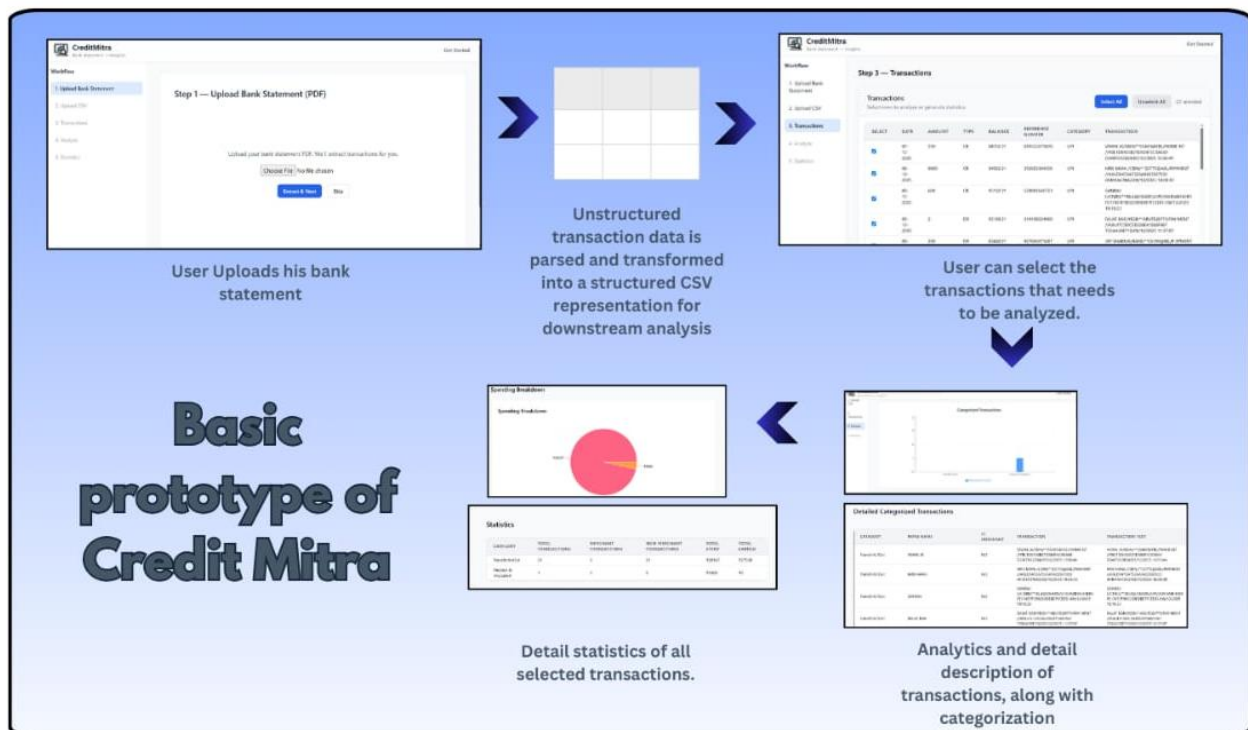


Fig. 2: Our preliminary developed prototype of CreditMitra

Sr. No.	Product / Solution Category	Definition	Qualifying Features / Differentiators
2	Improvements to existing Products with significant value add	Credit-risk models become more dynamic by incorporating merchant information. The entire pipeline ensures privacy, reduces merchant lookup costs through LangSearch and the LLM knowledge base, and minimizes reliance on third-party applications.	Reduced application cost, enhanced execution speed, higher credit-risk model accuracy, and improved loan processing and underwriting performance.

6 Deliverables.

1. A robust, fully validated pipeline for extracting information from diverse bank statements and transforming unstructured data into structured form.
2. A payee extraction model capable of handling varied transaction-string formats used by different banks.
3. A merchant/non-merchant classification model with high recall that identifies merchant transactions based on payee name.
4. Enhancing credit-risk scoring models by integrating real-time merchant information to make them more dynamic and responsive.
5. Adding privacy across the entire pipeline to prevent data leaks during data collection, processing, and model inference.
6. A scalable, distributed GUI-based application capable of serving numerous banks and NBFCs, offering an intuitive interface with functionalities such as bank-statement transaction extraction, merchant information retrieval, transaction categorization, credit-risk scoring, and a comprehensive analytics dashboard.

7 Project Plan with Major Milestones

Work Package 1: Dataset Collection and Pipeline Setup (1-2 months)

Collect diverse bank statements from multiple banks for testing and coverage. Build the initial pipeline for bank-statement extraction, including text extraction using Docling and transformation of unstructured text into structured transactions. Create a manually labelled

dataset for fine-tuning SLMs. Pipeline validation will include format testing across banks, field-level accuracy checks, consistency verification, and stress-testing on noisy/complex layouts to ensure robustness.

Work Package 2: Dataset Creation and SLM Fine-Tuning for Payee Extraction & Classification (2-3 months)

Manually create a high-quality dataset for fine-tuning an SLM to extract payee names from heterogeneous transaction strings. The model should learn to identify the payee entity regardless of the statement format or string pattern. Develop a second manually labelled dataset for merchant vs. non-merchant classification, ensuring coverage of diverse banks, narration styles, and rare transaction formats. Fine-tune the SLM for both payee extraction and classification tasks, focusing on robustness, high recall, and generalization across varied transaction patterns.

Work Package 3: Merchant Information Extraction Pipeline Using LLM Knowledge Bases (2-3 months)

Develop a robust pipeline for extracting merchant information using the knowledge base of various LLMs. Experiment with multiple LLMs to optimize the trade-offs between cost, knowledge coverage, and response speed. Implement a fallback mechanism using LangSearch when merchant information is unavailable or uncertain. The final system should generate a concise three-line merchant summary describing what the merchant does, its domain, and its relevance for credit-risk assessment.

Work Package 4: Dynamic Credit-Risk Modeling with Real-Time Merchant Information (1-2 months)

Develop dynamic credit-risk models that incorporate real-time merchant information to improve prediction accuracy and risk assessment. Integrate merchant insights directly into the scoring pipeline so the model adapts as new merchants emerge or customer behavior changes. Ensure strong privacy protections throughout model development and inference by incorporating differential privacy, on-device processing, and secure data-handling mechanisms.

Work Package 5: End-to-End Pipeline Testing, Optimization, and Debugging (1-2 months)

Test the complete pipeline across a wide variety of heterogeneous bank statements to evaluate robustness and ensure consistent extraction quality. Measure and improve the system's response speed and the accuracy of the credit-risk models through iterative refinement. Identify bugs, edge cases, and failure modes across modules including extraction, payee detection, merchant lookup, and scoring and resolve them to achieve production-grade stability.

Work Package 6: GUI Application Development, Merchant Storage, and System Scalability (1-2 months)

Develop an easy-to-use GUI application that supports bank-statement extraction, transaction analytics, and credit-risk scoring. Integrate a vector database for storing merchant embeddings to enable fast and accurate retrieval. Ensure the system is fully scalable through distributed components and maintain strong privacy protections by keeping sensitive processing local or within secure environments. The final application should provide a seamless end-to-end workflow for underwriting and analysis.



Fig. 4: Overall flow of the proposed project plan

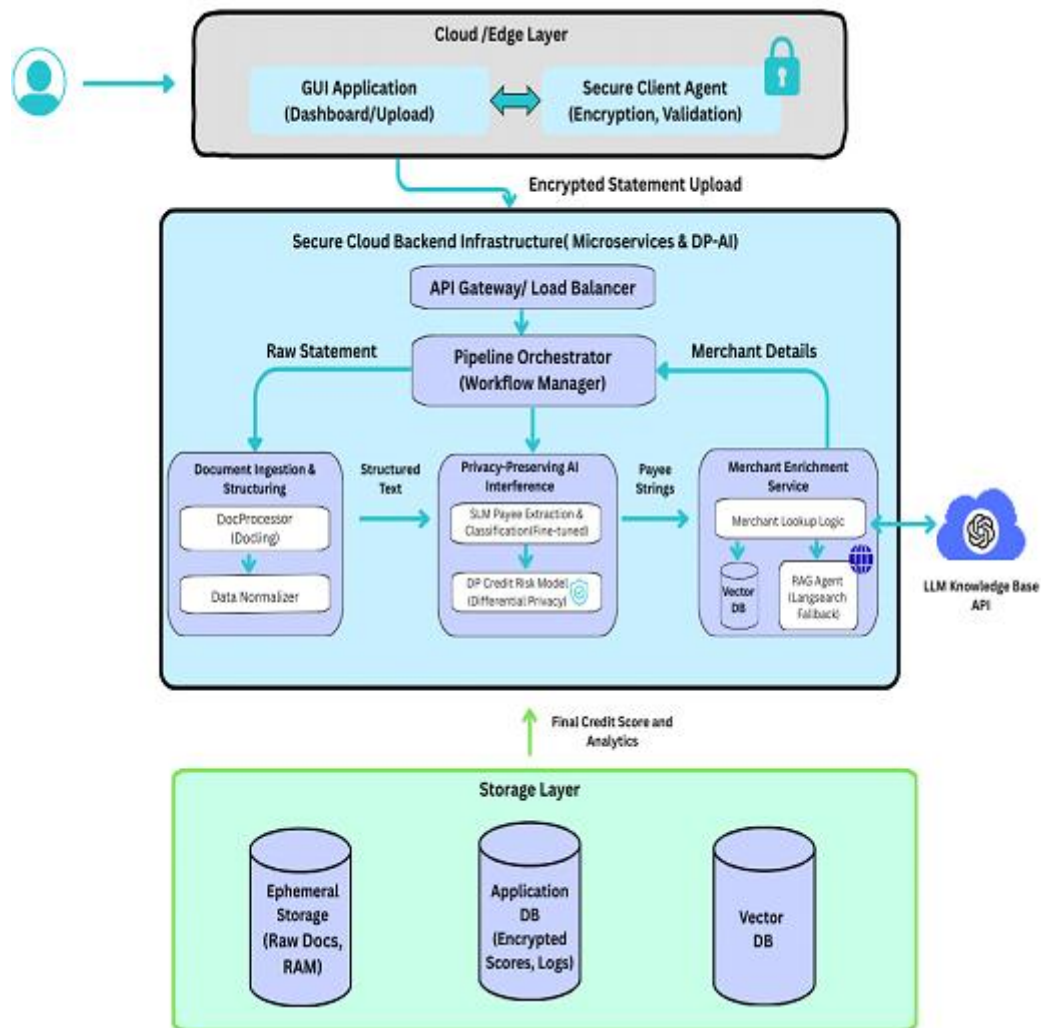


Fig. 3: Detailed architecture of CreditMitra

The detailed architecture of proposed CreditMitra incorporating data protection using differential-privacy is shown in Fig. 3.

Sr. No.	Milestone	Target Date	Remarks
1	Dataset and Pipeline Setup	1 month	Dataset for both payee name extraction model and merchant/ non-merchant identification model will be made.
2	SLM Fine-Tuning (Payee & Class)	2 months	Fine-tuned models (payee name extraction, and merchant/non merchant identification) will be privacy aware.
3	Merchant Extraction (LLM Knowledge base)	2 months	Fallback of Langsearch will be incorporated, information found about merchants will be stored in a vector database.
4	Dynamic Credit Risk Modelling		High focus will be on making models dynamic; by incorporating information about merchants will be there, the models are privacy aware
5	End to End Pipeline (Testing and Optimization)	2 months	The whole pipeline will be tested on vast variety of bank statements; the pipeline will be tested both at user level and bank level.
6	GUI App Storage and Scalability	3 months	Features will be incorporated in easy-to-use GUI apps, and prototypes will be given for testing and validation.

8 Challenges / Risks & Mitigation

- Dataset is not available to solve this problem, we need to use bank statements of different people to get the dataset, also merchant transactions are mostly only 10% of all the transactions, so dataset is highly imbalanced.
- Fine tuning SLM's take high computations, and datasets need to be labelled manually.

9 Resources and budget

- **GPU access is required for fine-tuning SLMs:** We already have GPUs/workstation in our lab.
- **A one-time LangSearch license cost of approximately ₹40,000** will be required.

S. No.	Budget Head	Justification	Total (INR)
1	Students Fellowship (UG)	@1000*10 months (for 3 UG Students)	300000
2	Contingency	@20000*3 (for 3 UG students)	60000
Total			3,60,000

Bibliography:

- [1] A. Trivedi, S. Mukherjee, R. K. Singh, V. Agarwal, S. Ramakrishnan, and Himanshu S. Bhatt. "TabSniper: Towards Accurate Table Detection & Structure Recognition for Bank Statements." *arXiv preprint arXiv:2412.12827*, 2024.
- [2] Glennon, Dennis, and Peter Nigro. "Evaluating the performance of static versus dynamic models of credit default: evidence from long-term small business administration-guaranteed loans." *Journal of Credit Risk* 7, no. 2, pp. 3-35, 2011.
- [3] <https://langsearch.com/>
- [4] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.T., Rocktäschel, T. and Riedel, S., "Retrieval-augmented generation for knowledge-intensive NLP tasks," *Advances in neural information processing systems*, 33, pp. 9459-9474, 2020.
- [5] Yu, D., Naik, S., Backurs, A., Gopi, S., Inan, H.A., Kamath, G., Kulkarni, J., Lee, Y.T., Manoel, A., Wutschitz, L. and Yekhanin, S., 2021. Differentially private fine-tuning of language models. *arXiv preprint arXiv:2110.06500*.
- [6] Yu, Manjiang, and Priyanka Singh. "Differentially Private Fine-Tuning of Large Language Models: A Survey," *International Conference on Advanced Data Mining and Applications*, Singapore", Springer Nature, pp. 100-113., 2025.